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RESEARCH

Evaluation of *Aquamicrobium lusatiense* NLF 2–7 as a Biocontrol Agent in Manure Composting: Effects on Odorous Compounds and Microbial Community Under Mesophilic Conditions

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Abstract

Microbial inoculation is a commonly applied approach in composting to enhance organic matter biodegradation and reduce odor emissions. However, the different characteristics of bacteria in terms of temperature can be considered to optimize their effect during different phases of composting. A mesophilic bacterium, namely *Aquamicrobium lusatiense* NLF 2–7, was evaluated to mitigate odor emissions and enhance the bacterial community under mesophilic composting. Two different treatments were designed: treatment 1 with a single inoculation on the initial day and treatment 2 with split inoculation at the initial and after 2 weeks. Results show that the treatments improve organic matter decomposition by 17.7–28.6% and significantly reduce volatile sulfur compound emissions, especially dimethyl sulfide (DMS) and hydrogen sulfide (H_2S) during the initial phase of composting. DMS emissions were mostly emitted in the first week, with reduction rates of 60.3% and 61.5% in both treatments, respectively. Additionally, mean phenol emissions were reduced by 7.9% in treatment 1 and 11.7% in treatment 2. The dominant bacterial phyla during composting were *Bacillota*, *Pseudomonadota*, *Bacteroidota*, and *Actinomycetota*, comprising 74 to 95% of the total population. This experiment suggests that *A. lusatiense* NLF 2–7, which is known for reducing sulfur emissions, can also enhance organic matter decomposition. Split inoculation appears more beneficial, with an initial inoculation managing sulfur emissions early on, followed by a second inoculation after the thermophilic phase to control phenol emissions throughout the composting process.

Keywords Manure composting · Odorous emission · Microbial additives · Bacterial dynamics

Introduction

Livestock manure generation has become an advanced environmental concern due to its potential to be a source of odorous compound emissions. Annually, 80–130 million tons of manure from total nitrogen is produced around the world, of which only 20–40% is utilized as fertilizer [1]. Animal manure is well known as a valuable source of organic nitrogen for plants. However, it can also be a source of environmental pollution due to odor emissions [2]. Furthermore, exposure to manure is linked to the proliferation of antimicrobial-resistant bacteria and genes, posing a serious threat to public health and environmental sustainability [3].

Composting is a long-established, cost-effective, and environmentally friendly method for managing animal manure, which is rich in biodegradable materials such as organic matter, nitrogen, phosphorus, and potassium [4, 5]. Commonly, the composting process is distinguished into 3 different phases, including mesophilic, thermophilic, and maturation [6]. The thermophilic phase is well known for rapid biodegradation, pathogen inactivation, and weed seed destruction [7, 8]. The mesophilic phase, which is identified with moderate temperature, has been known as the dominant

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phase in the composting process [9, 10]. Furthermore, this phase contributes as a critical window for odor generation. Odorous compounds are primarily emitted during the mesophilic and thermophilic phases [11]. Zhu et al. (2016) and Zheng et al. (2022) reported that most pollutants, including volatile sulfur compound emissions, are released between 43 and 83.4% during the mesophilic and pre-thermophilic stages [12, 13]. Hydrogen sulfide (H_2S), methyl mercaptan (MM), dimethyl sulfide (DMS), and dimethyl disulfide (DMDS) are volatile sulfur compounds (VSCs) that are commonly emitted in the composting of animal manure [8, 26].

This research focuses on *Aquamicrobium lusatiense* NLF2-7, a mesophilic bacterium that is commonly found in wastewater treatment plants [14]. *Aquamicrobium lusatiense* has a temperature growth range of 12–44 °C with optimal growth at 30–37 °C. It is also capable of thriving in a broad pH range of 4–10 [15]. By sustaining the composting process at a mesophilic temperature, we ensured that the *Aquamicrobium lusatiense* continued to be active throughout the process. In regular composting that reaches the thermophilic phase > 50 °C, mesophilic bacteria would be rendered inactive or even eliminated. Consequently, this research was designed to intentionally omit the thermophilic phase to preserve the viability and functionality of the *Aquamicrobium lusatiense*.

Previous experiments by Fritche et al. (1999) found the effectiveness of *Deflubacter lusatie*, which has been reclassified as *Aquamicrobium lusatiense* by Kampfer et al. (2009), to degrade chlorophenols and phenolic compounds [15, 16]. Additionally, genome analysis of strain NLF2-7 has revealed the presence of the SOX pathway (soxXYZABCD), which may facilitate the complete oxidation of sulfur atoms in thiosulfate to sulfate, potentially reducing sulfur emissions [14]. Based on the above literature, this study aims to investigate the impact of *A. lusatiense* NLF2-7 inoculation on odor emissions and bacterial composition during the composting of the swine and cattle manure mixture.

Materials and Methods

Bacterial Isolation

Strain NLF 2–7 was obtained from biofilm samples collected at livestock wastewater treatment facilities in Nonsan, Republic of Korea. Biofilms are organized microbial communities that attach to surfaces and are encased in a self-produced extracellular matrix. The samples were diluted and plated on sulfur-oxidizing bacteria medium (per liter: $Na_2S_2O_3 \cdot 5H_2O$ 8.0 g, K_2HPO_4 2.0 g, KH_2PO_4 2.0 g, NH_4Cl 0.4 g, $MgCl_2 \cdot 7H_2O$ 0.2 g, $FeSO_4 \cdot 7H_2O$ 0.01 g, agar 15.0 g) [17] and incubated aerobically at 30 °C for 10 days. Strain NLF 2–7 was then isolated, subcultured on tryptic soy agar medium, and preserved as purified colonies in a 15% (v/v) glycerol solution at − 80 °C for long-term storage. Strain NLF 2–7, identified as *Aquamicrobium lusatiense*, is a Gram-negative, non-flagellated, rod-shaped bacterium [14].

Odor Mitigation Efficacy During the Composting Process

Composting Materials

The composting mixture comprised swine manure, dairy manure, and sawdust at a 5:4:1 ratio. Table 1 shows the initial characteristics of the mixture, which had a moisture content of 74.3 ± 0.19%.

Experimental Design

The experiment was conducted using cylindrical reactors with a diameter of 14.7 cm and a height of 28.7 cm. Each reactor contained approximately 1.3 kg of compost mixture (Supplementary Fig. 1). The aeration rate was maintained between 0.3 and 0.4 L/min·kg-VS. Three experimental groups were established: a control group without microbial inoculation, treatment 1 with 1% (w.b.) microbial inoculum applied on the initial day (single inoculation), and treatment 2 with a 0.5% (w.b.) microbial inoculum applied on the initial day, followed by another 0.5% application two weeks into the composting process (split inoculation). A single colony was inoculated into 100 mL of Tryptic Soy Broth (TSB) and incubated at 30 °C with shaking at 150 rpm for 2 days. The resulting cultures were then adjusted to an optical density of 1.0 at 600 nm (OD_{600}) using a spectrophotometer, corresponding to an inoculation density of approximately 6×10^8 CFU/mL (colony-forming units). The compost was mixed weekly, and the temperature sensor was placed at the center of the composting material and monitored in real time using a Campbell Scientific CR1000 data logger connected to an AM16/32 relay multiplexer (Campbell Scientific Inc., Logan, US).

Table 1 Characteristics of the compost mixture (swine manure, dairy manure, and sawdust)

Characteristics	Value
Moisture (% w.b)	74.3 ± 0.19
Volatile solid (% d.b)	91.9 ± 0.15
Bulk density (kg/m^3)	0.78 ± 0.02
Free air space (%)	29.6 ± 0.05
C/N	18.2 ± 2.57

*Data represented as mean ± S.D, *w.b* wet basis, *d.b* dry basis, *n* = 3

Odor Gas Analysis

Gas samples were collected in Tedlar® bags (CEL Scientific Corp., CA, USA) to evaluate odor emissions. Ammonia (NH_3) emissions were monitored using a gas detection module (LGD Compact-A, Axetris, Kaegiswil, Switzerland) based on tunable diode laser absorption (TDLA) spectroscopy. Sulfur compounds (hydrogen sulfide (H_2S), methyl mercaptan (MM), dimethyl sulfide (DMS), dimethyl disulfide (DMDS)) were measured via gas chromatography (CP-3800, Varian Inc., CA, USA) equipped with a pulsed flame photometric detector (PFPD) and a capillary column (CP-Sil 5 CB, length: 60 m, inner diameter: 0.32 mm, film thickness: 5 µm). A thermal desorption instrument (UNITY series 1, Markes International Ltd., Bridgend, UK) was used for sample pretreatment before sulfur compound analysis by GC. VFAs, such as acetic acid, propionic acid, iso-butyric acid, butyric acid, iso-valeric acid, and valeric acid, along with VOCs, including phenol, p-cresol (4-methylphenol), indole, and skatole (3-methylindole), were assessed using chromatography (7890A, Agilent Technologies Inc., Santa Clara, CA, USA) equipped with a flame ionization detector (FID) and a capillary column (DB-Wax Ultra Inert, length: 30 m, inner diameter: 0.25 mm, film thickness: 0.25 µm). Prior to analyzing, VFAs and VOCs were absorbed onto glass and stainless-steel adsorption tubes containing mixed adsorbents (Carbopack C, Carbopack B, Carbopack X). The gases were absorbed using an integrating pump (MP-∑30KNII, Sibata Scientific Technology Ltd., Saitama, Japan) at a flow rate of 100 ml per minute for 5 min. CO_2 concentrations were measured using Vaisala CARBOCAP solid-state sensors (model GMT 221, Vaisala, Helsinki, Finland). Following each weekly mixing, odor intensity and odor unit measurements were conducted. Odor intensity was assessed using the dynamic-scale method according to ASTM International Standard E-544–99, with odor intensity referencing scales (OIRS) ranging from 10 points, correlating to 12 to 6200 ppm n-butanol equivalent. Odor units were measured with a Scentroid SM100i (IDES Canada Inc.) equipped with a U-A plate (2114 OUE—5.1 OUE).

Analysis of Bacterial Community

Compost samples were collected at 1, 2, 3, and 5 weeks during the composting process. Three samples from each group were pooled in equal proportions, immediately frozen at − 80 °C, and processed simultaneously for further analysis following standard procedures. Metagenomic DNA extraction from manure samples was performed using the DNeasy PowerSoil Kit (Qiagen, Hilden, Germany) according to the manufacturer's instructions. The V3–V4 regions of the 16S rRNA gene were amplified using the primers 341 F and 805R:

341 F; 5′-TCGTCGGCAGCGTCAGATGTGTATAAGAGACAGCCTACGGGNGGCWGCAG-3′; 805R;5′GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAGGACTACHVGGGTATCT AATCC-3′. The Quant-IT PicoGreen assay (Invitrogen, CA, USA) was employed for pooling and normalizing the amplified DNA products. Purification of the PCR products was carried out using AMPure beads, and the final purified samples were quantified via qPCR in accordance with the KAPA Library Quantification Protocol Guide (KAPA Library Quantification kits for Illumina sequencing platforms). Quality assessment was performed with the TapeStation D1000 ScreenTape system (Agilent Technologies, Waldbronn, Germany). All sequencing was conducted on an Illumina MiSeq platform (Illumina, San Diego, CA, USA) at Macrogen, Inc. (Seoul, Republic of Korea).

Data Analysis

The forward (Read1) and reverse (Read2) sequences were trimmed to 250 bp and 200 bp, respectively, and sequences with more than 2 expected errors were discarded. Batch-specific error models were created to reduce sample-specific noise, and the corrected paired-end reads were merged into single sequences. Chimeric sequences were then filtered out using the DADA2 consensus method, producing Amplicon Sequence Variants (ASVs). Additionally, to standardize the data for microbial community analysis, QIIME (v1.9) was employed to subsample and normalize read counts based on the sample with the lowest read count.

Taxonomic classification was conducted using BLAST + (v2.9.0) based on the organism with the highest sequence similarity, but assignments were excluded if the best hit had less than 85% query coverage or an identity score below 85% in the matched region. Additionally, multiple sequence alignments of ASV sequences were performed with MAFFT (v7.475), and a phylogenetic tree was generated using FastTreeMP (v2.1.10) [18].

The ASV abundance and taxonomy data were utilized for microbial cluster comparison analyses using QIIME [19]. Species diversity and evenness within microbial clusters were assessed by calculating the Shannon and Inverse Simpson indices. Alpha diversity was further analyzed through rarefaction curves and Chao1 estimates. Beta diversity across samples was evaluated using Weighted UniFrac distances, and the results were visualized through Principal Coordinates Analysis (PCoA) and UPGMA dendrograms. The data were organized using Microsoft Excel 2016, and the statistical significance of physicochemical characteristics and odor emissions between the control and both treatments was assessed based on *p*-values using SPSS 26.0 (IBM SPSS Inc., USA) and RStudio 4.2.3. Additionally, the Pearson correlation between selected physicochemical factors and odor emissions, as well as between the dominant phylum and odor emissions, was analyzed using RStudio 4.2.3.

Results and Discussion

Changes in Physicochemical Characteristics During Composting

Temperature and Moisture Content

Microorganisms rapidly break down the organic content of the compost substrate, raising the composting temperature. The temperatures increased to 45.7 °C, 45.2 °C, and 45.6 °C for the control, treatment 1, and treatment 2, respectively, shortly after composting started (Fig. 1A). During the experiment, the insulator was not installed to maintain the mesophilic conditions to facilitate adequate temperature for *Aquamicrobium lusatiense* NLF 2–7. Additionally, the relatively small amount of compost due to the reactor size helps to facilitate heat loss, preventing overheating during composting.

The moisture reduction rates, calculated based on the initial and final moisture mass at the end of composting, were 24.2% for the control group, 26% for treatment 1, and 28.7% for treatment 2. While no significant difference was found between the control and treatment 1 ($p > 0.05$), a significant difference was observed with treatment 2 ($p < 0.05$). This moisture loss is likely attributed to evaporation driven by the temperature increase during composting.

Carbon Dioxide, Volatile Solids, and C/N Ratio

CO_2 emissions are a biogenic product and mostly come from organic matter biodegradation. The high amount of readily degradable organic matter resulted in high CO_2 emissions during the initial stages of composting. 166.65 mg/day·g-VS for the control, 149.94 mg/day·g-VS for treatment 1, and 158.94 mg/day·g-VS for treatment 2 were produced on the initial day of the experiment. After peaking, CO_2 emissions decreased, and by day 27, CO_2 emissions dropped around 12 mg/day·g-VS, indicating that the compost had reached maturity (Fig. 1B). Correspondingly, there was a significant effect on volatile solid reduction compared to the control and

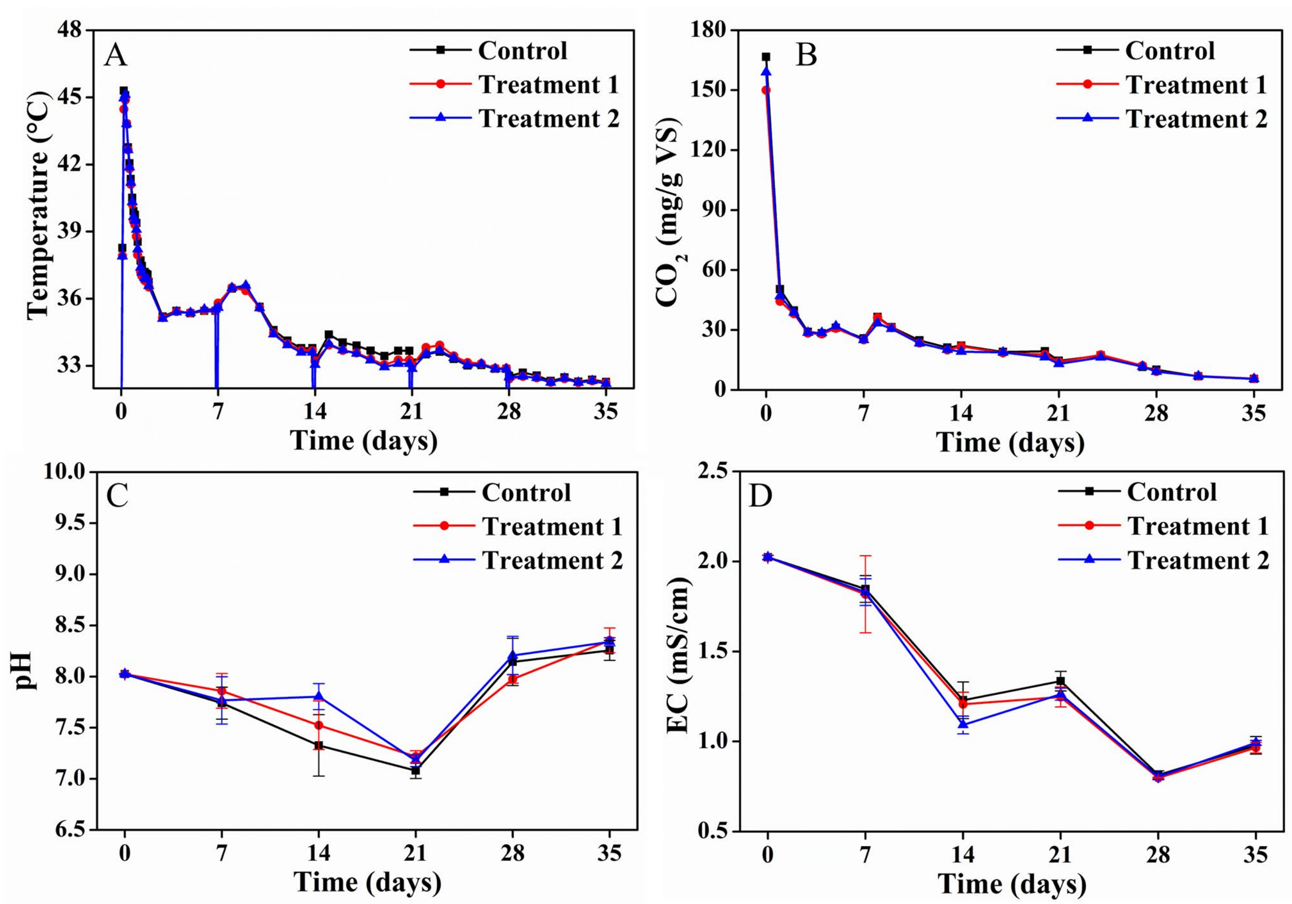


Fig. 1 Variations of temperature (**A**), daily CO_2 emissions (**B**), pH (**C**), and EC (**D**)

treatment groups ($p < 0.05$) (Table 2). Treatment 1 increases the organic matter decomposition by 17.7% while treatment 2 achieves 28.6% compared to the control group. Applying *Aquamicrobium lusatiense* can accelerate organic matter biodegradation to reach the maturity phase, which can minimize time and labor during the composting process. This finding proves and supports the results that some microbial additives can expedite organic matter degradation and mineralization during composting [20, 21]. The carbon-to-nitrogen (C/N) ratio decreased from 18.2 to 15.6 for control, 17.7 for treatment 1, and 14.8 for treatment 2 (Table 2). This reduction is due to a faster carbon breakdown than nitrogen mineralization, indicating nutrient availability and compost maturity [22].

pH, EC, Bulk Density, and Free Air Space

The pH pattern shows a suitable range for the composting process, providing a suitable environment for microorganisms [23]. pH was maintained in 7.1–9.4 at all treatment groups (Fig. 1C). Electrical conductivity (EC) significantly decreased from 2023 mS/cm to a range of 978.7 to 993.00 mS/cm across all treatments (Fig. 1D). An increase in bulk density was monitored from 0.78 g/cm^3 to 0.92–0.96 g/cm^3 in all experimental groups (Table 2). During composting, the particle size was decreased due to compaction and organic matter breakdown by microorganisms, leading to compaction and finally increased bulk density (Table 2). Consequently, the FAS decreased from 29.5 to 13.43–16.60% in all experimental groups (Table 2). The increase in bulk density and decrease in FAS were found common in composting studies, highlighting the impact of composting in reducing particle size, finally changing the bulk density and porosity [24].

Odor Mitigation Efficacy During the Composting Process

Ammonia Emissions

The NH_3 emissions reached the highest emission on the third day of composting (Supplementary Fig. 1 A). The NH_3 emissions generally increase during the early stages of composting, attributed to the high decomposition of nitrogen-based compounds [9, 25]. Table 3 displays the weekly and overall means of NH_3 emissions recorded during a 35-day composting interval. Notable variations in NH_3 emissions were detected at intervals throughout the 35-day composting duration. In the 0–7 day interval, the average NH_3 emissions were 62.26 μg/day·g-VS for the control group, 62.21 μg/day·g-VS for treatment 1, and 59.69 μg/day·g-VS for treatment 2. Treatment 2 exhibited a statistically significant reduction compared to the control ($p = 0.013$). However, NH_3 emissions decreased significantly after the first week, aligning with the findings of previous studies that reported nearly undetectable NH_3 emissions after two weeks due to the consumption of degradable organic matter and nitrogen fixation by microbes [26].

The control group exhibited an overall mean emission of 22.30 mg/day·g-VS, whereas treatment 1 and treatment 2 recorded emissions of 22.35 mg/day·g-VS and 21.31 mg/day·g-VS, respectively. Treatment 2 exhibited a slight reduction of approximately 4.4% ($p = 0.007$). These results indicate that dividing the inoculation into two times, even with the same number of microorganisms, was more effective in

Table 2 Changes in volatile solids, total carbon, total nitrogen, CN ratio, bulk density, and free air space during composting

Parameters		Control	T-1	T-2	SEM	*p*-value	
						T-1	**T-2**
VS reduction (g(%))		66.5 (21.8)	78.3 (25.3)	85.5 (27.7)	3.33	0.04	0.03
TC (%, d.b.)	**Initial**	45.50	45.50	45.50			
	Final	44.26	45.19	44.80	0.35	0.13	0.31
TN (%, d.b.)	**Initial**	2.53	2.53	2.53			
	Final	2.88	2.62	3.02	0.10	0.23	0.18
C/N ratio	**Initial**	18.19	18.19	18.19			
	Final	15.56	17.67	14.82	0.73	0.18	0.20
Bulk density (g/cm^3)	**Initial**	0.78	0.78	0.78	0.00	1.00	1.00
	Final	0.96	0.94	0.92	0.01	0.44	0.40
Free air space (%)	**Initial**	29.51	29.60	29.60	0.00	0.28	0.23
	Final	13.43	15.34	16.60	0.01	0.42	0.26

*Data represented as means ± SEM with different letters in the same column are significantly different ($p < 0.05$); *T-1* treatment 1, *T-2* treatment 2, *VS* volatile solid, *TC* total carbon, *TN* total nitrogen, $n = 3$

*The VS percentage reduction was calculated relative to the initial VS mass

Table 3 Mean emission and percent reductions of NH_3 and DMS

Emission (µg/day/g-VS)	Time	Mean emission (µg/day·g-VS)			SEM	Reduction		*p*-value	
		Control	**T-1**	**T-2**		**T-1**	**T-2**	**T-1**	**T-2**
NH_3	**0–7 (*n* = 21)**	62.26	62.21	59.69	0.57	0.07	4.12	0.989	0.013
	8–14 (*n* = 15)	2.00	1.90	1.64	0.13	5.27	17.94	0.824	0.350
	15–21 (*n* = 9)	0.01	0.01	0.02	0.00	37.83	− 65.38	0.421	0.254
	22–28 (*n* = 9)	0.02	0.01	0.01	0.00	44.56	24.47	0.380	0.615
	29–35 (*n* = 6)	0.04	0.03	0.02	0.00	40.29	48.34	0.019	0.017
	Overall (*n* = 60)	22.30	22.25	21.31	0.21	0.20	4.44	0.496	0.007
DMS	**0–7 (*n* = 9)**	1.30	0.52	0.50	0.14	60.33	61.51	0.013	0.016
	8–14 (*n* = 9)	0.98	1.11	1.24	0.08	− 12.91	− 26.03	0.479	0.295
	15–21 (*n* = 9)	0.05	0.03	0.02	0.01	44.02	62.13	0.133	0.050
	22–28 (*n* = 9)	0.02	0.06	0.08	0.01	− 214.80	− 311.27	0.028	0.182
	29–35 (*n* = 6)	0.0018	0.0017	0.0017	0.00	5.83	6.10	0.891	0.892
	Overall (*n* = 42)	0.48	0.41	0.45	0.03	14.88	7.27	0.141	0.444

**SEM* standard error of mean, *T-1* treatment 1, *T-2* treatment 2, *n* number of measurements

reducing NH_3 emissions. NH_3 emissions during composting are significant due to their contribution to air pollution and the loss of nitrogen. Multiple factors affect NH_3 emission, including composting material type, C/N ratio, temperature, moisture content, aeration, and microbial composition [27, 28]. In all experimental groups, the temperature profile peaked sharply after composting began, contributing to the high NH_3 volatilization on the first day. The NH_3 release is closely linked to the pile's temperature and the activity of microorganisms [28, 29]. Ammonification is essential in the nitrogen cycle, where organic nitrogen substances such as proteins are converted into NH_3. Through enzymatic activities, microorganisms, mainly bacteria and fungi, break down these nitrogen-rich substances, producing NH_3 as a byproduct. This process, which occurs during composting and decomposition, facilitates nitrogen cycling. NH_3 can subsequently be nitrified to generate nitrates, essential for plant growth.

Sulfur Compounds

H_2S, MM, DMS, and DMDS are prevalent sulfur compounds generated during the composting of animal manure [11, 30]. This study demonstrated the absence of MM and DMDS during the 35-day composting process. H_2S was identified solely on the initial day, with emission recorded at 0.50 µg/day·g-VS in the control, while treatment 1 emitted 0.13 µg/day·g-VS, and treatment 2 emitted 0.11 µg/day·g-VS (Supplementary Fig. 2B). Treatment 1 and treatment 2 exhibited statistically significant reductions compared to the control ($p < 0.05$).

The low H_2S emissions can be ascribed to several parameters, such as oxygen availability, pH, microbial activity, material composition, and the conversion of sulfur components into alternative sulfur compounds [31–33]. Since DMS was the primary sulfur compound identified in this investigation, it indicates that most sulfur elements were converted into DMS via methylation. The highest DMS emissions occurred during the first week of composting (Supplementary Fig. 2C), showing a notable rise on day 7, followed by a sharp decrease to minimal levels. This pattern corresponds with Zhao et al. (2019), who reported fast DMS generation during the initial 5 days of municipal sludge composting [34]. Table 3 displays the weekly and overall average DMS emissions. In the 0–7 day interval, the control group released 1.30 µg/day·g-VS of DMS, whereas treatment 1 and treatment 2 released 0.52 µg/day·g-VS and 0.50 µg/day·g-VS, respectively. Both treatments demonstrated statistically significant decreases relative to the control ($p < 0.02$). During the 8–14 day interval, the average DMS emissions increased to 1.11 µg/day·g-VS for treatment 1 and 1.24 µg/day·g-VS for treatment 2, in contrast to 0.98 µg/day·g-VS for the control. Treatment 1, which involved a single inoculation on the initial day, may have demonstrated a greater reduction in DMS emissions than treatment 2 due to its effectiveness during the first and second weeks of composting, when emissions peaked and DMS levels were at their highest.

The overall DMS emissions were 0.41–0.48 µg/day·g-VS in all experimental groups. Treatment 1 resulted in a 14.88% reduction, while treatment 2 achieved a 7.27% reduction. The results demonstrate that whereas both treatments significantly reduced DMS emissions in the first week, the total reduction in DMS emissions throughout the full composting duration was minimal. DMS, a sulfur molecule, greatly influences odor during composting and possesses a minimal olfactory detection threshold of 0.0083 mg/m^3 [35]. The DMS emissions are primarily affected by the sulfur concentration in the composting material, microbial activity, and the breakdown of organic matter [11, 30]. The observed decrease in DMS emissions during the preliminary composting phase is likely attributable to microbial sulfur metabolism, mainly via the methylation of H_2S to produce DMS. The genomic investigation of the *A. lusatiense* NLF 2–7 strain utilized in the treatments identified the SOX pathway (soxXYZABCD), which enables the oxidation of sulfur compounds, specifically the transformation of thiosulfate to sulfate [36].

Volatile Organic Compounds

Volatile organic compounds, such as phenol, p-cresol, indole, and skatole, were observed throughout the composting process. Nonetheless, only phenol emissions were observed during the 35-day composting duration. Daily phenol emissions throughout 35 days exhibited an early peak, subsequently followed by a progressive decrease (Supplementary Fig. 3A). Table 4 displays the average phenol emissions at various time intervals. In the 0–14 day interval, the control group released 0.54 µg/day·g-VS of phenol, whereas treatment 1 and treatment 2 exhibited decreased emissions of 0.50 µg/day·g-VS and 0.49 µg/day·g-VS, respectively. Treatment 2 demonstrated a significant reduction of 9.48% ($p = 0.01$), showing its efficacy in phenol reduction during the early stage of composting.

During 15–35 day intervals, phenol emissions significantly decreased across all experimental groups. The control group emitted 0.37 µg/day·g-VS, while treatment 1 and treatment 2 showed emissions of 0.33 µg/day·g-VS and 0.32 µg/day·g-VS, respectively. Overall phenol emissions were 0.45 µg/day·g-VS for the control group, 0.42 µg/day·g-VS for treatment 1, and 0.40 µg/day·g-VS for treatment 2, yielding overall reductions of 7.9% for treatment 1 and 11.7% for treatment 2 ($p < 0.02$). The results indicate that treatment 2 consistently exhibited greater efficacy in reducing phenol emissions across all phases of the composting process. Giving inoculation in two doses at different times, while maintaining the same quantity of microorganisms, enhanced microbial activity during composting, consequently facilitating a more efficient breakdown of phenolic compounds.

The reduction of phenol may be attributed to microbial activity, particularly phenol-degrading bacteria such as *Aquamicrobium lusatiense*, which have been demonstrated to play a significant role in the degradation of phenolic compounds [15, 16]. Phenol is an aromatic chemical predominantly obtained from the decomposition of lignin, a fundamental structural component in plant tissue. The degradation of lignin during composting produces phenolic monomers, which are beneficial in the formation of humic compounds [37]. While these compounds contribute to humic acid synthesis, elevated phenol concentrations might negatively impact microbial activity, hindering the breakdown process due to their antibacterial capabilities [38]. Phenol may hinder vital microbial enzymes, altering metabolic pathways needed for the decomposition of organic matter and nutrient cycling. Increased phenol concentrations in compost might adversely affect plant growth when utilized as a soil amendment.

Volatile Fatty Acids

The primary VFAs detected in this study were acetic acid and propionic acid, which supports the findings of Plachá et al. (2013) [39]. Other volatile fatty acids, including isobutyric acid, butyric acid, isovaleric acid, and valeric acid, were identified solely on the initial day of composting (Supplementary Fig. 3B-G), suggesting that these compounds are

Table 4 Mean emission and percent reductions of phenol and VFA

Emission	Time (day period)	C	T-1	T-2	SEM	Reduction (%)		*p*-value	
						T-1	**T-2**	**T-1**	**T-2**
Phenol (µg/day·g-VS)	**0–14 ($n = 15$)**	0.54	0.50	0.49	0.01	6.73	9.48	0.05	0.01
	15–35 ($n = 15$)	0.37	0.33	0.32	0.01	9.60	14.91	0.01	0.02
	Overall ($n = 30$)	0.45	0.42	0.40	0.01	7.90	11.70	0.02	0.01
Acetic acid (µg/day·g-VS)	**0–14 ($n = 15$)**	0.080	0.086	0.114	0.01	− 7.74	− 43.55	− 7.0	0.03
	15–35 ($n = 15$)	0.049	0.041	0.047	0.00	17.75	4.99	17.9	0.34
	Overall ($n = 30$)	0.065	0.063	0.081	0.00	2.03	− 24.95	− 2.5	0.03
Propionic acid (µg/day·g-VS)	**0–14 ($n = 15$)**	0.016	0.016	0.015	0.00	− 1.05	3.71	0.96	0.53
	15–35 ($n = 15$)	0.032	0.029	0.025	0.00	10.39	21.87	0.64	0.08
	Overall ($n = 30$)	0.024	0.022	0.020	0.00	6.64	15.91	0.60	0.07
Total VFA (µg/day·g-VS)	**0–14 ($n = 15$)**	0.109	0.114	0.139	0.01	− 3.87	− 27.26	0.94	0.05
	15–35 ($n = 15$)	0.081	0.069	0.072	0.00	14.87	11.61	0.07	0.10
	Overall ($n = 30$)	0.095	0.091	0.106	0.00	4.13	− 10.67	0.87	0.06

**SEM* standard error of mean, *C* control, *T-1* treatment 1, *T-2* treatment 2, *n* number of measurements quickly consumed or converted throughout the composting process. Acetic acid constituted more than 65% of the total VFA emissions in each group, followed by propionic acid at over 14%, with the remainder including other VFAs, such as isobutyric, butyric, isovaleric, and valeric acids. The average total VFA emissions throughout the composting process, derived from 30 measurements, were 0.095 μg/day·g-VS for the control, 0.091 μg/day·g-VS for treatment 1 (a 4.13% reduction), and 0.106 μg/day·g-VS for treatment 2 (a 10.67% increase). Nonetheless, the variations were not statistically significant ($p > 0.05$).

VFAs are essential intermediates generated during composting, generally as a result of fermentative respiration in hypoxic conditions. Elevated levels of VFAs are recognized to induce phytotoxicity and suppress microbial activity. Their undissociated states, particularly under acidic conditions, pose a threat to microorganisms [40]. Acetic, propionic, butyric, and isobutyric acids are mostly produced by the metabolism of easily degradable compounds, such as carbohydrates, lipids, and proteins. Greater molecular weight volatile fatty acids, such as valeric and isovaleric acids, are typically associated with protein degradation [39]. This study revealed that VFA emissions reached their peak in the initial stages of composting, attributed to the fast breakdown of easily degradable organic substances. This effect occurs when biodegradable materials are subjected to anaerobic conditions in the compost pile.

Odor Intensity and Odor Unit

Supplementary Fig. 4 demonstrates the fluctuations in odor intensity and odor units throughout a 35-day composting period. The control group started with an odor intensity of 3.5, which progressively diminished to 1.5 by day 35. Treatment 1, which began with a slightly elevated odor intensity compared to the control group, exhibited a consistent decrease, ending at 1.4 by day 35. Treatment 2 exhibited a more consistent decreasing trend, averaging approximately 1.3 for the composting duration. The overall downward pattern across all groups indicates that composting significantly mitigates odors over time.

The control group reported the greatest odor units on the first day, measuring 1.040 OUE, whereas treatment 1 and treatment 2 started with lower values of 945 and 805 OUE, respectively. Throughout the composting process, all experimental groups exhibited a significant reduction, with odor units decreasing to 12 OUE by day 35. Despite the lack of statistical significance in odor unit variations among the groups ($p > 0.05$), the significant decrease from the start to the final days illustrates the efficacy of composting in odor reduction. The control group exhibited a reduction of 57.8%, whereas treatments 1 and 2 attained reductions of 63% and 62.4%, respectively. Odor emissions from composting pose considerable environmental and public health issues due to the discharge of diverse malodorous compounds. Research indicates significant relationships between odor intensity and primary odorants, including NH_3, sulfur compounds, and VOCs [31, 41]. Zhu et al. (2016) found DMS as an important odorant in sewage sludge compost, whereas Li et al. (2021) revealed a logarithmic correlation among NH_3, DMS, H_2S, and odor levels in kitchen waste composting [42]. Zhao et al. (2019) similarly emphasized the contributions of NH_3 and volatile sulfur compounds to odor production in the initial phases of municipal sludge composting [34]. In this study, the reductions in H_2S, DMS, and phenol indicate that microbial inoculation may effectively mitigate odor emissions during composting.

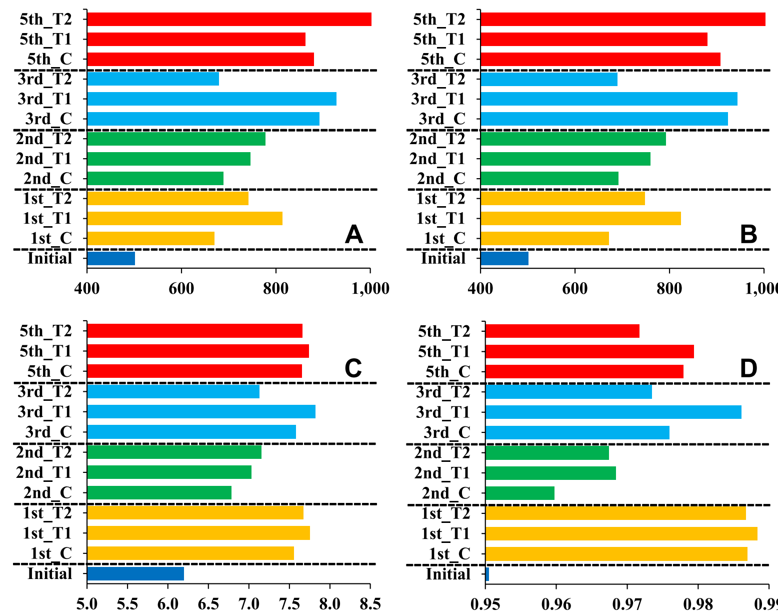
Analysis of Bacterial Community

Figure 2 presents the alpha diversity of all groups. The number of ASVs was higher in the treatment groups (813 and 741) compared to the control group (670) on the 1st week, and this trend persisted until 2nd week of the composting process. Subsequently, the ASV numbers fluctuated and finally had similar values, particularly control and treatment 1 (880 and 862, respectively). Treatment 2 shows the highest ASV with a value of 1010. Similar findings have been reported in another experiment involving the addition of thermotolerant nitrifying bacteria in sewage sludge composting that could enlarge the ASVs to 932 compared to the control (875) [43].

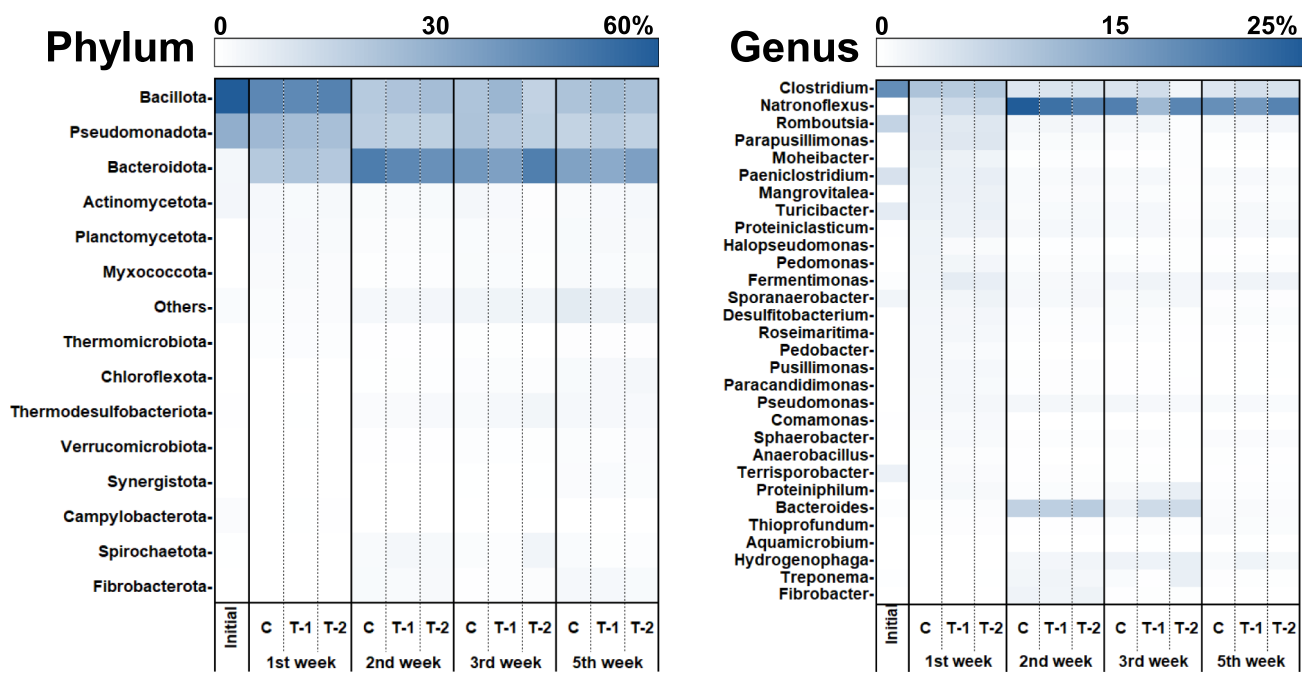
The coverage index for all composting samples ranged from 0.95 to 0.99, signifying that the sequencing depth adequately captured bacterial diversity. The Chao1, Shannon, and Gini-Simpson indices were employed to evaluate the richness, diversity, and evenness of the bacterial population. The results show that treatment groups had higher microbial richness, evenness, and diversity throughout the composting process, as evidenced by the Chao1 and Shannon indices, in comparison to the control group. The treatment groups had higher Chao1 values than the control group on the 1st and 2nd weeks. On the final day, treatment 2 shows the highest value. In the case of Shannon, treatment 2 exhibited elevated values on the 1st week, 2nd week, and final day. The Gini-Simpson index demonstrated that all groups had high diversity throughout the composting processes. These findings are supported by other research that shows the microbial addition enhanced Chao1 and Shannon indices during the composting process [43, 44].

The bacterial communities throughout all experimental groups and several sampling points (initial, 1st, 2nd, 3rd, and 5th week) demonstrated dynamic fluctuations in relative abundance during the composting process (Fig. 3 and Supplementary Table 1). The predominant bacterial phyla among all groups were *Bacillota*, *Pseudomonadota*,

Fig. 2 Alpha diversity of the bacterial community in different groups during composting: ASVs (**A**), Chao1 (**B**), Shannon (**C**), and Gini-Simpson (**D**)



*C: Control; T1: Treatment 1; T2: Treatment 2; ASVs: Amplicon sequence variants



*) T-1: Treatment 1, T-2: Treatment 2

Fig. 3 Relative abundance of top 15 phyla and genera during composting

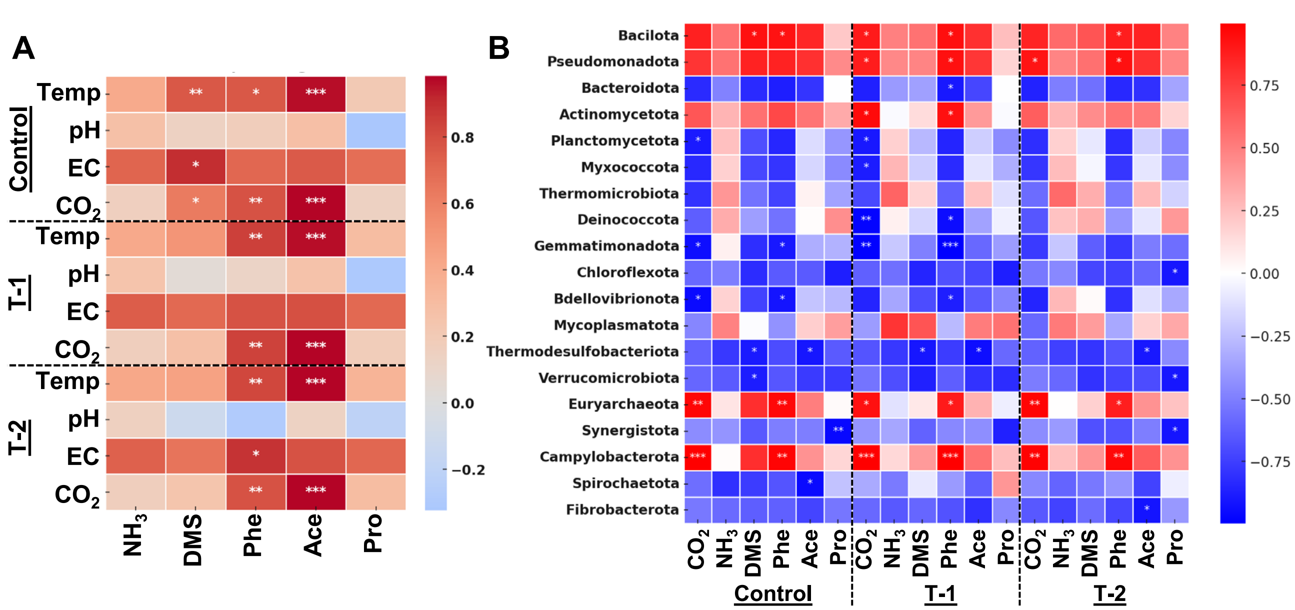
Bacteroidota, and *Actinomycetota*, collectively comprising 74 to 95% of the overall bacterial population. This finding was supported by other research that shows the high abundance of these bacterial phyla during composting [45, 46]. Initial samples showed the dominance of *Bacillota* and *Pseudomonadota* (total 90%) among all treatments. On day 7, *Bacillota* continued to be the dominating phylum, even though its proportions decreased to 43.64%, 43.00%, and 45.58% in the control, treatment 1, and treatment 2, respectively. *Pseudomonadota* also shows the decreasing trends. Conversely, *Bacteroidota* had an increasing trend and showed an abundance of 19.81–21.8%

in all experimental groups. The decreasing trends of the dominant phylum imply the effectiveness of composting to enhance bacterial diversity. Furthermore, the three predominant phyla, *Bacteroidota*, *Bacillota*, and *Pseudomonadota*, are well known in composting for their capability in organic matter decomposition [47–50]. Minor phyla, such as *Planctomycetota* and *Myxococcota*, emerged during this stage, indicating early microbial diversification.

The 1st week, 2nd week, and final day exhibited a predominance of *Bacteroidota*, including 34.01%, 31.03%, and 34.89% in control, treatment 1, and treatment 2, respectively. *Myxococcota* and other minor phyla, including *Synergistota* and *Fibrobacterota*, demonstrated a modest increase from days 14 to 35, indicating their involvement in the later stages of composting and their contribution to the degradation of more complex materials such as cellulose [51]. *Thermodesulfobacteriota*, a thermophilic sulfate-reducing phylum, was also detected at a later phase and may correlate with sulfur reduction [52]. Additionally, *Fibrobacteria* that play roles in fiber decomposition were detected in the 2nd week and on the final day of composting. The genus level was dominated by *Clostridium*, *Natronoflexus*, *Romboutsia*, *Parapusillomonas*, and *Bacteroide*s. *Clostridium* demonstrated a high abundance after the first week of the composting process (9–11%) but decreased to 4–5% on the final day. Conversely, *Natronoflexus* demonstrated an increasing trend after the first week forward.

The correlation heatmap was examined to evaluate the correlations between chosen physicochemical parameters and odor emissions (Fig. 4A). While not statistically significant, direct positive correlations were noted between NH_3 and temperature, pH, and CO_2 in all treatment groups. The correlation between DMS emissions and physicochemical parameters varied between the treatment groups and the control. Notably, no substantial correlations were detected in the treatment groups, suggesting a possible role of *Aquamicrobium lusatiense* in influencing microbial dynamics modulating DMS emissions. Furthermore, emissions of phenol and acetic acid demonstrated direct positive relationships with temperature, electrical conductivity, and carbon dioxide across all treatments.

The heatmap analysis demonstrated clear correlations between microbial taxa and all treatment groups (Fig. 4B). *Bacillota* demonstrated a high positive correlation with DMS in the control group, with CO_2 in treatment 1, and with phenol across all groups. *Pseudomonadota* correlates with both CO_2 and phenol in both treatment groups. *Bacteroidota* showed a negative correlation with phenol in treatment 1, but *Actinomycetota* showed a positive correlation with phenol in the same treatment. *Myxococcota* and *Deinococcota* exhibited a substantial negative correlation with CO_2 in treatment 1. *Chloroflexota* had a negative correlation with propionic acid in treatment 2, but *Verrucomicrobiota* demonstrated a negative correlation with DMS in the control group. *Thermodesulfobacteriota* has a negative correlation



*) *represents $p<0.05$, ** represents $p<0.01$, represents $p<0.001$; T-1: Treatment 1, T-2: Treatment 2, Temp: Temperature, Phe: Phenol, Ace: Acetic acid, Pro: Propionic acid

Fig. 4 The correlation heatmap between: **A** selected physicochemical factors and odor emissions; **B** dominant phylum and odor emissions with DMS in the control and treatment 1, as well as a pronounced negative correlation with acetic acid across both groups. Moreover, *Euryarchaeota* and *Campylobacterota* consistently exhibited a direct positive correlation with both CO_2 and phenol in all experimental groups. These findings highlight the diverse microbial patterns of odor and physicochemical characteristics among experimental groups, suggesting potential functional roles in the observed environmental conditions.

Odor-mitigating bacteria were identified, including families within *Pseudomonadota*, such as *Alcaligenaceae* (NH_3 elimination), *Dietzia* (VFA degradation), and *Comamonas* and *Aquamicrobium* (sulfur compound reduction). While not predominant, these groups played a role in the initial reduction of malodor [53, 54] (Supplementary Table 2). Sulfur emissions, including H_2S and DMS, alongside phenol, lead to odor problems and possible health risks, requiring solutions for their reduction. Recent studies have demonstrated the effectiveness of various composting methods in diminishing sulfur emissions and phenol levels. The data indicate that the timing and treatment application are essential elements in optimizing the composting process.

The potential applications of microbial additives for odor mitigation can be extended to real-world composting practices. *A. lusatiense* NLF 2–7 could be applied during the initial composting phase in windrows, static piles, or in-vessel composting systems. For instance, in large-scale livestock farms that produce substantial amounts of manure, the microbial solution may be directly sprayed onto the composting materials at the beginning of the process. Subsequently, split inoculation can then be performed during the turning operations, which are standard practices in windrow composting. Such applications have the potential to reduce specific odorants, including sulfur-based compounds and phenol, thereby improving air quality and minimizing environmental complaints.

Conclusions

This study demonstrates that *Aquamicrobium lusatiense* NLF 2–7 reduces DMS and phenol emissions during the composting of swine and cattle manure mixture under mesophilic conditions, showing notable effectiveness in reducing DMS during the early stages and in reducing overall mean phenol emissions. Organic matter decomposition improves by 17.7–28.6%, which is important for accelerating the composting process and minimizing time consumption. Moreover, split inoculation led to greater decomposition compared to single inoculation. The observed changes in the microbial community suggest that microbial additives can enhance bacterial diversity. Although inoculation is necessary at the start of composting to reduce DMS, which is primarily emitted during the early stages, the compost typically enters a thermophilic phase within the first week. Since *A. lusatiense* NLF 2–7 is a mesophilic microorganism, it is likely to become inactivated under high-temperature conditions. Therefore, applying a second dose after the thermophilic phase, as the compost enters the mesophilic maturation stage, can help further reduce phenol emissions that occur throughout the composting process. For this reason, split inoculation may be more advantageous than single inoculation. Since this study was conducted at the laboratory scale and evaluated *A. lusatiense* NLF 2–7 exclusively under mesophilic conditions, future research should focus on scaling up the inoculation process to evaluate the effectiveness of split dosing throughout a complete composting cycle, including the transition into thermophilic conditions. Furthermore, additional studies are needed to assess the cost-effectiveness, application strategies, and adaptability of *A. lusatiense* NLF 2–7 across diverse composting feedstocks and environmental conditions.

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Author Contributions R.W. contributed to investigation, data analysis, writing, visualization, validation, software development, and data curation. J.S. contributed to investigation and formal analysis. S.L. contributed to validation, formal analysis, and investigation. J.L. contributed to investigation and formal analysis. Y.H.N. contributed to resources, formal analysis. M.H.L. contributed to supervision and funding acquisition. K.I.H. contributed to conceptualization, data curation, validation, visualization, and review. H.A. contributed to investigation writing, review, editing, supervision, project administration, methodology, data curation, conceptualization, and funding acquisition. All authors reviewed the manuscript.

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Data Availability This whole genome shotgun project has been deposited at DDBJ/ENA/GenBank under the accession JAKSED000000000, BioProject PRJNA807442, and BioSample SAMN25981959. The Sequence Read Archive raw reads are deposited under accession numbers SRR18547513 and SRR22071623.

Declarations

Competing Interest The authors declare no competing interests.

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